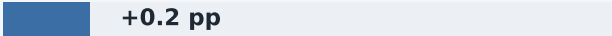


Folding hallmark	Vision — Δ top-1 accuracy	Language — Δ perplexity
C1 Energy minimisation <i>Thermodynamic folding funnel</i>	Langevin descent on $E_\theta(z x)$  +0.2 pp	Energy-gated attention +60.9 PPL <i>(Langevin refinement: no effect)</i>
C2 Adaptive topology <i>Native contact formation</i>	Multi-head attention over particles (once at init)  +1.3 pp	Standard per-layer self-attention — <i>subsumed by standard Transformer</i>
C3 Cooperativity <i>Local $\rightarrow$ global coupling</i>	Depthwise-separable 1D convolution  +0.5 pp	Feed-forward network — <i>subsumed by standard Transformer</i>
C4 Chaperone <i>Stress-gated assisted folding</i>	$\ \nabla_z E\ $ -gated corrective module  +0.4 pp	Intra-block pre/post-FFN comparison +4.9 PPL
C5 Environmental sensitivity <i>Cellular-context modulation</i>	FiLM-style context scale & shift  +0.4 pp	Layer normalisation — <i>subsumed by standard Transformer</i>

Contributions are the performance drop when each module is removed from the full model (Table: cross-domain decomposition). Vision: CIFAR-10 Δ top-1 accuracy. Language: WikiText-2 Δ perplexity.